Signaling, screening or sunk costs? Experimental evidence on how prices affect agricultural technology adoption in Uganda

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Abstract

Free samples are a widely used strategy to introduce new products or technologies, offering prospective users the opportunity to gain first-hand experience and potentially facilitate diffusion through social networks. However, concerns remain that giving away products for free may reduce their perceived value, increasing the risk that recipients will under-utilize, repurpose, or resell the product rather than use it for its intended purpose. We explore three mechanisms through which charging a positive price may increase uptake, intended use and subsequent adoption of a new technology: (1) a signaling effect, where a positive price conveys unobservable product attributes such as expected benefits and quality; (2) a screening effect, whereby payment deters users who do not value the product and targets those more likely to use it; and (3) a sunk cost effect, where paying a positive price induces a psychological commitment to use. We test how these pricing mechanisms shape uptake, use, and subsequent adoption of a recently released maize seed variety, drawing on a field experiment with smallholder farmers in Uganda. We find that willingness to pay is a reliable predictor of subsequent use of seed trial packs, pointing to the value of modest prices for targeting likely adopters. We find no evidence that the initial offer price conveyed quality signals affecting farmer behavior. At the same time, sunk cost effects are negative, suggesting that charging farmers can reduce the quality of experimentation. While adoption rates in the subsequent season do not differ across treatment

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arms, screening effects on production persist, most likely because the majority of farmers recycle seed from their trial harvest and pricing-induced management differences affect recycled seed quality. These findings carry important implications for how pricing strategies can be designed to promote technology adoption in low-income settings.

Keywords: technology diffusion, trial packs, screening, sunk cost effect, signaling.

1 Introduction

Prices are ubiquitous in economic transactions. They are central to the efficient allocation of scarce resources within a society and provide important supply-side incentives. But charging the (full) price may not always be optimal. For instance, if a product or technology is new, providing it for free or at a discount for a short period of time may be necessary to encourage potential users to try it and learn from it (Bawa and Shoemaker, 2004). From a social welfare point of view, subsidies may be justified to ensure access to essential goods and services for poor or disadvantaged communities that may benefit most (Duflo and Banerjee, 2011). Additionally, providing an initial bundle of subsidized goods or services can help overcome barriers to adoption by giving a temporary boost that enables reinvestment and sustained use over time, potentially setting off a path to long-term welfare gains, in a process reminiscent of the dynamics described in the poverty trap literature (Balboni et al., 2021; Visser et al., 2020). More broadly, from a policy perspective, in the early stages of market development, public subsidies can play a catalytic role by stimulating initial demand for promising technologies or services, helping them reach a critical mass of users that enables economies of scale and attracts sustained private sector engagement. Finally, the presence of positive externalities provides another strong rationale against charging the full price for socially beneficial goods and services (Miguel and Kremer, 2004).

At the same time, there are concerns that free or subsidized provision of goods and services can diminish their perceived value among recipients. When this occurs, such goods may remain unused, be repurposed, or even resold. For instance, free bed nets have sometimes been used for fishing, and subsidized chlorine for water treatment has occasionally been diverted to household cleaning (Cohen and Dupas, 2010; Ashraf et al., 2010). This tension is especially salient in the design of public health and agricultural interventions, where concerns about cost recovery, wastage, and behavioral responses to subsidies often shape policy.

There are at least three mechanisms through which charging a positive price for a good or service may increase its uptake and intended use. First, prices can act as *signals* of unobserved product attributes, such as quality, effectiveness, or expected benefits. For example, in contexts of uncertainty and asymmetric information, consumers may infer higher quality from higher prices, increasing the extent to which they value and use the product (Milgrom and Roberts, 1986;

Bagwell and Riordan, 1991). Second, a *screening* effect may arise: individuals who place a higher intrinsic value on the product are more likely to be willing to pay for it, leading to better targeting of actual users. Third, a *sunk cost* effect refers to a psychological commitment whereby individuals are more likely to value and use a product simply because they have incurred a cost to obtain it, regardless of its objective utility (Kahneman and Tversky, 1979; De Bondt and Makhija, 1988).

A key challenge is that these mechanisms are typically conflated in observational data, as each mechanism implies a positive correlation between price and use. Building on Ashraf et al. (2010) who employ a two-stage pricing design in which commodities are first offered at randomized prices and then subject to a surprise discount to disentangle screening and sunk cost effects, we extend this design by introducing an additional initial stage that allows us to also isolate signaling effects. Specifically, in the first stage, farmers receive an initial randomized offer price that provides a potential signal of unobservable product attributes. In the second stage, a bargaining game elicits willingness to pay, which allows us to identify screening effects. In the third stage, a surprise discount is introduced, enabling us to measure sunk cost effects. The contribution of each mechanism on a particular outcome can then be estimated through regression analysis: To capture signaling effects, we assess the impact of the initial randomized offer price while controlling for both willingness to pay and whether the farmer paid for the good or service. To identify the screening effect, we focus on the marginal effect of willingness to pay while controlling for the initial randomized offer price and whether the farmer ultimately paid or received the good or service for free. Finally, to test for sunk cost effects, we focus on the indicator for full payment versus free provision, controlling for willingness to pay and the initial offer price.

We apply our experiment in the context of seed trial packs distributed to smallholder farmers in a developing-country setting with the aim of promoting the use of seed of recently introduced improved crop varieties. Studying how pricing may affect adoption and use of such agricultural technologies presents a policy-relevant case for several reasons. First, given the novelty of and limited familiarity with a new variety, trial packs may be necessary to encourage experimentation among risk-averse farmers (Foster and Rosenzweig, 1995; Brick and Visser, 2015). Second, social learning, especially through peer networks, plays a critical role in the large-scale adoption of new agricultural technologies, suggesting the existence of significant positive externalities (Conley and Udry, 2010; Bandiera and Rasul, 2006; Van Campenhout, 2021). Third, seed purchases involve non-trivial upfront costs, making them susceptible to sunk cost effects (Soman, 2001). Finally, seed markets are often plagued by significant information asymmetries regarding product quality, which adds an additional layer of risk to farmer investment decisions and gives rise to potential price signaling (Bold et al., 2017; Ahmad, 2022). For instance, Bulte et al. (2023) show that in Kenya, uncertainty about seed quality reduces labor investment, and argue that the presence of low-quality ('lemon') inputs undermines learning about profitability, while bundling seed of improved crop varieties with crop insurance to reduce uncertainty has been found to increase farmer effort and learning (Bulte et al., 2020). Miehe et al. (2023) provide evidence that, in Uganda, agro-input dealers actively use signaling to counter these asymmetries and build trust with buyers.

The experiment was implemented in Uganda with about 760 smallholder farmers. To assess how pricing of improved maize seed varieties affects outcomes along the adoption pathway, we collected detailed data in two rounds: in the short run (the first season after distribution) we tracked whether the seed trial pack was planted, whether seed and harvest were kept separate, who in the household used it, and whether complementary inputs such as fertilizer and chemicals were applied; in the longer run (the subsequent season) we measured adoption of improved and promoted varieties, complementary input use, and production on a randomly selected plot. This dual focus enables us to go beyond the immediate use of the trial pack and examine whether pricing effects persist into subsequent agricultural decisions.

We find that paying for the seed reduces rather than reinforces appropriate use, contrary to the sunk cost hypothesis. Farmers who received the seed for free were more likely to keep the harvest separate, a behavior that facilitates learning and evaluation. At the same time, willingness to pay is positively associated with more appropriate use of the trial seed, greater investment in complementary inputs, and stronger intentions to adopt improved varieties (including the promoted seed), suggesting that prices may screen for farmers who are better positioned to benefit from new technology. We find no evidence that the initial offer price served as a signal for unobserved product attributes. While adoption rates in the subsequent season do not differ across treatment arms, screening effects on production persist: since most farmers recycle seed from their trial harvest, pricing-induced differences in first-season management appear to affect the quality of recycled seed and thus subsequent production. Taken together, these results reveal a tension in pricing: while willingness to pay identifies farmers who use the trial seed more appropriately, requiring payment reduces the quality of experimentation. This suggests that eliciting willingness to pay may be valuable as a targeting tool, even if the seed is ultimately provided at a discount or for free.

Our study contributes to a growing literature that examines how pricing affects the uptake and adoption of goods and services. Seminal studies by Ashraf et al. (2010) and Cohen and Dupas (2010) offer contrasting insights. Ashraf et al. (2010), in a two-stage pricing experiment of chlorine for water purification in Zambia, find evidence of screening effects (suggesting that willingness to pay reflects private information about future use) but no support for sunk cost effects. Their results imply that pricing may improve targeting efficiency, potentially justifying lower subsidy levels. In contrast, Cohen and Dupas (2010) study demand and usage of insecticide-treated bed nets in Kenya and find that even small prices substantially reduce uptake, with little evidence that pricing improves targeting through screening mechanisms. They conclude that free distribution may yield greater health benefits by expanding coverage. More recently, Mahmoud (2024) uses a two-stage pricing experiment for improved wheat variety seed in Bangladesh to examine whether prices can screen farmers based on their returns to the technology. Farmers first face randomized subsidy levels, after which non-buyers are randomly offered the seed for free. This design allows a comparison between self-selected non-buyers and the broader population. The study finds that farmers who initially decline to purchase the seed adopt it when offered for free and achieve realized returns similar to those of the average farmer. These results suggest that willingness to pay does not reliably screen farmers based on returns and that prices may primarily restrict access rather than target the technology to higher-return users.

Our study builds on this body of work but adds an important innovation by explicitly isolating and testing signaling effects as a third mechanism alongside the more commonly studied screening and sunk cost effects. Existing two-stage pricing designs typically examine whether prices screen for users with higher private valuations or induce sunk cost-driven commitment, typically by estimating one effect while controlling for the other. However, if prices also signal product attributes in settings characterized by experience goods or information asymmetries, then ignoring this channel may confound estimates of both screening and sunk cost effects.[1] To address this limitation, we extend the canonical two-stage design into a three-part pricing experiment that separately identifies signaling, screening, and sunk-cost effects by randomizing the initial offer price (signaling), eliciting willingness to pay through a bargaining game (screening), and assigning a surprise discount to a randomly selected subsample of participants (sunk-cost effects).

A second contribution is the specific context of Uganda's maize seed market, where persistent concerns about counterfeit and low-quality seed in the supply chain undermine farmers' trust in new varieties (Barriga and Fiala, 2020; Bold et al., 2017). This makes signaling effects particularly relevant, while the importance of maize as both a food and cash crop ensures that screening and sunk cost mechanisms also play a role. The Ugandan setting thus provides a particularly informative context for examining how pricing strategies shape technology adoption.

The remainder of this article is organized as follows. We first explain the methods used and the experimental design, followed by the estimation strategy. We then turn to sampling and provide descriptive statistics of our study population. The analysis consists of two parts. We first look at the price elasticity of demand for the seed trial packs using willingness-to-pay data. We then provide estimates for screening, sunk cost, and signaling effects. The last section concludes and reflects on implications for policy.

[1] The reason is that signaling shifts willingness to pay; if this is not explicitly controlled for, willingness to pay absorbs part of the signaling effect and becomes endogenous. As a result, the coefficient on willingness to pay (interpreted as screening) will be biased and, because sunk-cost estimates are identified from payment outcomes conditional on willingness to pay, bias carries over to the sunk-cost effect estimate. Both Ashraf et al. (2010) and Cohen and Dupas (2010) discuss signaling as a theoretical possibility, but their design primarily focuses on disentangling screening from sunk cost effects.

2 Methods and experimental design

The standard approach to disentangle screening and sunk cost effects arising from charging a price is the *two-stage pricing design* pioneered by Ashraf et al. (2010) and Cohen and Dupas (2010). In a two-stage pricing design, the first stage involves randomizing the posted price of a product across participants or sites, generating variation in who initially decides to purchase. This allows researchers to test for screening effects, since willingness to buy at a given posted price may correlate with expected use. In the second stage, among those who agreed to purchase at the posted price, a subset is randomly offered an unanticipated discount or rebate that lowers the actual transaction price. Because all of these individuals revealed a willingness to pay the original posted price, any systematic difference in subsequent usage between those who ultimately pay more versus less can be attributed to sunk cost effects, rather than to differences in underlying demand. Together, this two-stage design cleanly separates selection at the offer stage from psychological responses to the act of paying.

We build on the standard two-stage pricing design by introducing an additional stage at the beginning, yielding a *three-stage pricing design*. In this additional first stage, farmers are presented with a randomized initial offer price. When a technology is new and its performance uncertain, such prices may be interpreted as signals about otherwise unobserved product attributes, such as quality, effectiveness, or expected benefits. Standard two-stage designs implicitly assume that price variation affects behavior only through screening and sunk-cost mechanisms. By randomizing the initial offer price prior to eliciting willingness to pay, our design allows us to test whether prices also influence behavior through signaling effects that shape farmers' beliefs about the technology. The second and third stages then follow the conventional design: we elicit willingness to pay (WTP), providing the basis for identifying screening effects, and finally, we grant a full (100%) surprise discount to a random subset of farmers, allowing us to test for sunk cost effects.

The experiment is implemented as a scripted bilateral bargaining game involving concessional offers. Farmers are offered the opportunity to purchase a bag of seed from a trained enumerator who follows a standardized bargaining protocol designed to mimic the types of negotiations commonly observed in real-world seed transactions.[2] The enumerator follows a standard script that was

[2]In two-stage pricing designs, such as those used by Ashraf et al. (2010) and Cohen and Dupas (2010), in the first stage, participants are typically offered a good at randomly assigned price points. A common drawback of this approach is that a substantial share of participants may opt not to purchase the good, which, absent over-subscription, can significantly reduce the effective sample size. Since willingness to pay (WTP) is central to analyzing two-stage pricing designs, alternative elicitation methods like the Becker-DeGroot-Marschak (BDM) mechanism can be employed to mitigate this problem to some extent. In its basic form, BDM asks participants to state a bid, which is then compared to a randomly drawn price. If the bid is lower, participants do not receive the product; if it meets or exceeds the price, participants purchase the product at the drawn price. A key advantage of this method is that it inherently includes a surprise discount for those whose WTP exceeds the random price—mimicking the second stage in two-stage pricing designs—see appendix E in Berry et al. (2020). While we initially planned to elicit WTP using a BDM mechanism, field testing revealed that farmers

implemented in Open Data Kit (ODK) on Android tablets.

As part of our first stage, an initial ask price was randomly drawn from a uniform distribution, and presented to the farmer as the price of one bag of seed for a new improved variety.[3] The enumerator explains what kind of variety it is and what the advantages are.

We then elicit willingness to pay (the second stage). The farmer has the option to accept the bag of seed at this initial offer price or not.[4] If the farmer does not accept the initial offer price, the farmer enters into bargaining where they are encouraged to name their first counter bid price. A computer algorithm then determines a counter-offer that the enumerator asks in a second round of negotiation. This new ask price is determined as the farmer's bid price plus 80 percent of the difference between the (initial) ask price and the farmer's bid price (appropriately rounded). This updated (lower) ask price is then compared to the bid price of the farmer. If the price difference between the bid and (updated) ask price is smaller than a (crop specific) threshold, the ODK script instructs the enumerator to sell at the price the farmer bids. If the difference between the updated ask price and the bid price is larger than the threshold, the updated ask price is presented to the farmer, and the farmer gets a second opportunity to accept or reject. If the farmer does not accept, they are encouraged to make a second bid and a third ask price is determined as the farmer's last bid price plus 80 percent of the difference between the last ask price and the farmer's last bid price. Bargaining continues until the farmer accepts an ask price, or the price difference between the last bid and last ask price is smaller than a (crop specific) threshold.[5]

A natural question is whether the bargaining game elicits true willingness to pay in the same way a BDM mechanism would. In a BDM, truthful revelation is a dominant strategy: participants have no incentive to misreport their WTP regardless of the randomly drawn price. In bargaining, by contrast, the farmer will start below their true WTP in early rounds, in an effort to strategically shade their WTP (Hortaçsu et al., 2018). However, the multi-round alternating-

were confused by the one-shot nature of the transaction. As a result, we opted for a bargaining game that more closely reflected farmers' real-world experience with price negotiation. This design change led us to update the pre-analysis plan, which is reflected incommit dated Feb 11, 2023: "changed PAP after field testing - dropped BDM for bargaining experiment" on GitHub.

[3]For maize seed packs (1kg) in Uganda, initial prices ranged from 9,000 to 12,000 Uganda shillings (UGX) in increments of UGX 1,000.

[4]In our Uganda sample, 118 out of 758 farmers in the seed arm (15.6 percent) accepted the initial offer without bargaining. For these farmers, willingness to pay is right-censored at the offer price: we know their WTP is at least as high as the initial offer, but not how much higher. This also means that, for this subgroup, the screening and signaling channels are collinear, as the negotiated price equals the randomly assigned initial price. These observations are retained in the analysis with their WTP set to the accepted offer price, which attenuates the screening coefficient toward zero and makes our screening estimates conservative. In an online appendix, as a robustness check, we re-estimate all main specifications after excluding farmers whose willingness to pay was right-censored. Results are similar to those reported here.

[5]To make the bargaining also incentive compatible for the enumerators, we told them in advance that the money that is collected from farmers during this first stage will be distributed equally among all the enumerators.

offer structure of our protocol resolves this concern. The final agreed price (whether the farmer accepts the last ask or the algorithm accepts the farmer's last bid) reflects the point at which the farmer was no longer willing to go higher, and therefore approximates the farmer's maximum WTP. To the extent that the negotiated price falls short of true WTP because some farmers capture surplus by stopping short of their maximum, this introduces classical measurement error that attenuates our estimates toward zero but does not bias the identification of screening, signaling, or sunk-cost channels.

After bargaining was concluded, we asked some additional control questions and, near the end of the interaction, offered a randomly selected subgroup of farmers a surprise discount (the third stage). Unlike most two-stage pricing designs that introduce random or tiered discounts (often to determine optimal subsidy levels), we employ a single, full (100%) discount. This decision is guided by four considerations. First, we anticipate a behavioral discontinuity between paying any positive amount and receiving the good for free, consistent with findings in behavioral economics that zero prices can have a disproportionate impact on uptake and use (eg. Shampanier et al., 2007; Shukla et al., 2022). Second, concentrating the sample in two distinct groups—those who pay and those who receive the good for free—maximizes statistical power for estimating the behavioral difference between paying and receiving the good, relative to designs that spread observations across many price points. Third, by replicating how new agricultural technologies are often introduced in practice, with free sample packs provided to encourage trial and learning, our approach strengthens the external validity of the study.[6] Fourth, our design focuses on a single, full discount rather than multiple rebate tiers, ensuring that all variation in whether farmers paid or not is generated directly by randomized assignment. This design isolates the sunk-cost effect at the extensive margin (pay versus free) in a transparent way. By contrast, canonical two-stage pricing designs identify sunk-cost effects from variation in the prices paid among those who choose to purchase. Because purchase itself is endogenous, separating sunk-cost effects from screening requires additional assumptions. Our design avoids this issue by generating exogenous variation directly at the pay-versus-free margin.

Our experimental design generates three sources of variation that allow us to identify distinct price mechanisms. First, the randomized initial offer price may act as a signal of otherwise unobserved product attributes, such as usefulness or quality. Second, the willingness-to-pay (WTP) price, defined as the final price agreed upon during the bargaining process, captures farmers' willingness to pay and thus provides the basis for screening. Third, we use an indicator of whether the farmer paid the agreed price or received the pack for free to identify the sunk-cost effect.[7]

[6] In Ashraf et al. (2010), the possibility of non-linearities around a zero price was also suggested by a practitioner (and tested in Panel B of their Table 4), indicating that such effects are not merely a theoretical curiosity but arise in real-world implementation contexts.

[7] As an alternative indicator closer in spirit to Ashraf et al. (2010), we also express this variable as the transaction price—equal to zero under the full discount and equal to the WTP price otherwise. Supplementary regressions using this specification are reported in the

Seed trial packs used in this study contain 1 kilogram of improved maize variety seed. We use a recently introduced hybrid variety popularly known as Bazooka, produced by Naseco Seed Ltd. The seed is high yielding. It promises between 3.5 and 4 metric tons per acre and was partly chosen because it is widely available on the market. The 1 kg bag of maize seed is enough to plant about one-eighth of an acre.

Our study sample constitutes 10 farmers per village from 76 villages in Uganda. The study is embedded in a larger factorial design comprising 148 villages in total. Our initial design called for 20 farmers per village across these villages. However, during piloting, we found that information about the bargaining opportunity spread quickly within villages, prompting us to reduce to 10 farmers per village. To compensate, we increased the total number of villages from 110 to 148, of which 76 were assigned to the seed treatment arm reported here (the remaining villages received alternative interventions not analyzed in this paper). Given an intra-cluster correlation of 0.10--0.15 for our primary outcomes (estimated from prior data on 3,450 smallholder maize farmers across 345 villages) this reallocation preserved effective statistical power, as in cluster-randomized designs, additional villages contribute independent variation, while additional farmers within the same village provide diminishing information due to within-village correlation. Simulation-based power calculations in the pre-analysis plan confirm comparable power before and after this design change.

Assignment to surprise discounts was randomized at the village level, as we wanted to avoid that a farmer receiving a bag of seed for free would have close interactions with farmers paying a positive price for the same trial pack. However, we did vary the initial offer price in the bargaining experiment at the individual level. Because the discount assignment varies at the village level, we cluster standard errors at the village level in all regressions.

3 Estimation

To separate the three different effects, we estimate regression models that are similar to the original two-stage design used in for instance Ashraf et al. (2010). Recall that in two-stage designs, study participants are given the opportunity to buy a commodity at different price points and in a second stage are given a surprise discount (leading to a proxy for the price the farmer is willing to pay and transaction price at which the farmer obtained the seed after the rebate). In such designs, the outcome variable of interest (for example, an indicator for whether the seed was used as intended) is regressed on both WTP price and the transaction price. A statistically significant and positive coefficient on WTP price, controlling for the transaction price, provides evidence of a screening effect, since it reflects the positive relationship between the participant's valuation and the outcome of interest, irrespective of whether the full price was paid or not. A significant and positive coefficient on the transaction price, controlling

appendix for robustness.

for the WTP price, indicates a sunk cost effect, where paying a higher price increases commitment to using the product, irrespective of the farmer's valuation of the product.

In our main analysis, we depart from the canonical specification in two ways. First, we add a signaling effect, identified from the randomized initial offer price, which may convey information about unobservable seed attributes such as the quality of the seed. Second, we identify the sunk cost effect using a discontinuous treatment indicator that equals one if the farmer paid the bargained price and zero if the farmer received the full surprise discount. This choice stays closest to the experimental design, where sunk costs were introduced through a randomized, sharp contrast between "free" and "paid" packs. By relying on this exogenous discontinuity, we avoid attributing sunk cost effects to within–paid-group variation in transaction prices that was jointly determined by bargaining and may therefore be endogenous. The interpretation is also straightforward: it captures whether paying anything at all changes subsequent use relative to receiving the pack for free.[8]

In our analysis, we thus estimate the following equation:

$$Y_i = \alpha + \beta_I I_i + \beta_P P_i + \beta_D T_i + \varepsilon_i \tag{1}$$

where Y_i is the outcome of interest (e.g., use of seed from trial pack, adoption of promoted variety in subsequent season,...), I_i is the initial offer price, P_i is WTP price, and T_i is a dummy variable indicating if the full price was paid. A statistically significant positive β_I coefficient in equation 1 provides evidence of a signaling effect ($\beta_I > 0$ in equation 1). Evidence of a screening effect is provided by a statistically significant and positive coefficient on WTP ($\beta_P > 0$ in equation 1). Finally, a statistically significant positive coefficient on the dummy variable indicating if the full price was paid provides evidence of a sunk cost effect ($\beta_D > 0$ in equation 1).[9] Because the surprise discount was randomized at the village level, we cluster standard errors at the village level throughout.

Our design, based on a fully randomized 100% discount, isolates the sunk cost effect at the extensive margin by comparing farmers who pay their reservation price with those who receive the seed for free. This maximizes statistical power and allows a direct test of the behavioral discontinuity at zero price. By contrast, designs that assign fixed discounts across multiple tiers estimate sunk

[8] A broader caveat applies to multistage pricing designs. In such designs, the "free" comparison group consists of participants who were first exposed to a price and only subsequently received a surprise discount, rather than participants who received the product without any prior price interaction. Because all participants pass through an initial pricing stage, price exposure itself may influence perceptions and subsequent use. Estimates of sunk cost effects in two-stage designs should therefore be interpreted as the effect of payment conditional on price exposure, rather than as the total effect of paying relative to pure free provision.

[9] Note that, in principle, these three tests can be derived from estimating a single equation (Equation 1). However, in the empirical section below we estimate three separate equations, each focusing on the specific effect of interest, and include the other independent variables demeaned and fully interacted with the variable of interest, following the regression adjustment method of Lin (2013), which improves precision and guards against covariate imbalance in randomized experiments.

cost effects along a continuous price gradient, which can be easier to interpret alongside the screening effect—also measured at the margin—and additionally provide insight into how sunk cost effects evolve with payment size rather than only at the free-versus-paid threshold. For completeness, in Appendix A we also present estimates based on the continuous transaction price, which stays closer to the canonical Ashraf et al. (2010) two-stage framework. In particular, we estimate Equation 1 again, but instead of T_i being a dummy variable, we now include the actual transaction price, which is zero if the discount was obtained, and equal to the WTP price if no discount was obtained. Note though that models using the actual transaction price in the context of an all-or-nothing discount are, after controlling for WTP and the randomized offer, effectively reparameterizations with the continuous coefficient mostly rescaling the paid-vs-free contrast by the conditional mean paid price.

In the analysis, we look at different (potentially correlated) outcomes, leading to the problem of multiple hypothesis testing. To deal with this problem, we follow the inverse-covariance weighted (ICW) index method proposed by Anderson (2008) and aggregate different outcome measures within broadly defined families into single summary indices. Each index is computed as a weighted mean of the standardized values of the outcome variables, with the weights derived from the inverse variance covariance matrix of the components of the index. Because the components are standardized to have zero mean and unit variance, a regression coefficient on the index can be interpreted as the effect in units of weighted standard deviations of the component outcomes.

4 Sample and Descriptive Statistics

The sample includes approximately 760 maize-growing households, selected to be representative of farmers across four districts in Eastern Uganda: Mayuge, Kamuli, Iganga, and Bugiri. These districts were selected because maize serves as an important crop for both food and income. In these four districts, 76 villages were randomly chosen from a complete list of villages, with the probability of selection proportional to village population size. From each selected village, 10 households were then randomly sampled.

Baseline data collection and the implementation of the experiment took place at the same time in February and March 2023, shortly before the start of the first agricultural season of 2023. The baseline survey asked questions about general household characteristics and more specific questions about farming and seed use. The data was collected by trained enumerators with at least three years of field experience, using standardized protocols that included an introduction and informed consent. Prior to the enumerators' visit for the bargaining experiment, farmers were informed that a team would visit their village and that there might be an opportunity to purchase a product, so they were encouraged to have a small amount of cash available.

Table 1 presents descriptive statistics and balance test results for our Uganda maize sample. The first column reports overall sample means; columns two and

Table 1: Descriptive Statistics and Balance

	Overall	Not discounted	Discounted	Difference	N
Age household head (years)	48.759	48.024	49.495	1.47	744
	(13.53)	(14.067)	(12.949)	(1.089)	
Head finished primary education	0.519	0.534	0.504	-0.03	759
	(0.5)	(0.499)	(0.501)	(0.042)	
Male household head	0.784	0.818	0.749	-0.069*	759
	(0.412)	(0.386)	(0.434)	(0.034)	
Household size	8.167	8.537	7.797	-0.74*	759
	(3.818)	(4.169)	(3.395)	(0.342)	
Distance to agro-dealer (km)	3.812	3.734	3.891	0.158	729
	(4.014)	(4.46)	(3.505)	(0.586)	
Perceives seed quality as important	0.415	0.432	0.398	-0.033	759
	(0.493)	(0.496)	(0.49)	(0.036)	
Uses promoted variety	0.063	0.068	0.058	-0.01	759
	(0.244)	(0.253)	(0.234)	(0.02)	
Seed from formal source	0.35	0.347	0.354	0.006	759
	(0.477)	(0.477)	(0.479)	(0.046)	
Buys seed often	0.195	0.216	0.174	-0.042	759
	(0.396)	(0.412)	(0.38)	(0.03)	
Plot size (acres)	1.11	1.141	1.078	-0.063	758
	(0.865)	(0.916)	(0.812)	(0.078)	
Yield (kg/acre)	421.851	407.313	436.349	29.036	749
	(359.59)	(364.235)	(354.789)	(29.932)	

Notes: Column 1 reports the overall sample mean with standard deviation in parentheses. Columns 2 and 3 report means with standard deviations in parentheses for the non-discounted and discounted groups. The Difference column reports the OLS coefficient on the discount indicator with village-clustered standard error in parentheses. Rows 1–5 are household characteristics; rows 6–11 are agricultural variables referring to the season prior to the baseline. $^{+}$ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$.

three show separate means for the non-discounted and discounted groups; and the Difference column tests whether the village-level discount randomization produced comparable groups. Household heads are on average nearly 50 years old, and about 52 percent have completed primary education. Average household size exceeds eight members. Farmers live, on average, roughly 4 kilometers from the nearest agro-dealer, indicating a relatively dense input supply network.

The two groups are well balanced on most covariates. Household size and the share of male-headed households show statistically significant differences, which is not unexpected given the number of covariates tested.

While the first five characteristics are expected to be unaffected by our intervention, we also report six baseline variables that are more likely to change as a result of the study. To begin, we include a broad measure of technology adoption by asking whether farmers had used seed of any improved maize variety (hybrid seed or open pollinating variety) on any plot during the season prior to the baseline. About 41 percent reported using improved maize varieties.

To capture uptake of the specific varieties promoted in our study, we took a more targeted approach. Farmers were asked to list all plots on which the relevant crop was grown, after which one plot was randomly selected for detailed follow-up questions.[10] Only around 6 percent of farmers had already planted the promoted maize variety, Bazooka, in the preceding season. This underscores that the technology introduced in the study had not yet gained traction at scale.

About 35 percent of farmers used maize seed from a formal seed source (for instance, from an agro-input dealer or the government extension system as opposed to own saved or shared between farmers) in the season preceding the baseline. To maintain sufficient vigor, seed, especially hybrid seed, should not be recycled too often. Fewer than 20 percent of maize farmers used seed that had been recycled fewer than five times, which is generally considered the upper limit for open-pollinated varieties. Finally, average maize yields on the randomly selected plot in the season preceding the intervention were around 400 kilograms per acre.

A midline survey was conducted in July 2023, after the first growing season, to capture use of the seed trial pack, agronomic practices, satisfaction with the trial variety, yields, and planning intentions for the next season. The second growing seasons started around August 2023 and harvesting took place in December or January 2024. An endline survey followed in February 2024 to measure actual adoption decisions, complementary input use, and production outcomes in the subsequent season.

5 Analysis

5.1 Price elasticity of demand

A first step in the analysis is to assess whether demand for the seed is responsive to price, since some degree of price sensitivity is a prerequisite for the screening channel. From a policy perspective, the magnitude of the price elasticity of demand indicates how sensitive farmers' purchasing decisions are to price changes and, therefore, how higher prices might constrain initial seed uptake independently of any behavioral mechanisms that influence subsequent use and adoption. In settings where farmers face cash or credit constraints, high price elasticity may disproportionately exclude those who could potentially benefit most from the technology. In that sense, high price elasticity is often cited as a key justification for offering subsidies. It is important to note that price elasticity captures how many farmers purchase the seed, whereas screening captures how those who purchase differ in their subsequent use.

Figure 1 shows the distribution of prices at which farmers agreed to purchase the maize seed trial pack during the bargaining experiment. We observe a negative relationship between price and the share of farmers who agree to

[10]The decision to only ask detailed questions on one (randomly selected) plot was guided by the fact that outcomes at plot level (such as adoption of improved inputs and technologies and production outcomes) are likely to be correlated within farm households such that gains in statistical power from surveying all plots likely do not outweigh costs of longer and more tedious questionnaires. As the plot to be surveyed was selected randomly, outcomes should be unbiased and consistent.

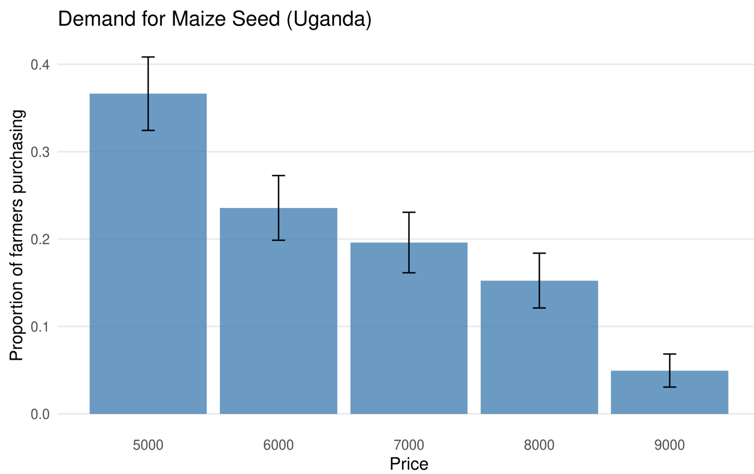

Figure 1: Distribution of WTP prices. The figure shows the share of smallholder maize farmers (N=758) in Uganda who agreed to purchase the seed trial pack at each final negotiated price during the bargaining experiment. Prices are final agreed prices after bargaining, in Uganda Shillings (UGX).

purchase the seed, consistent with a downward-sloping demand curve. Uptake declines steadily as the price increases, indicating relatively high price sensitivity. This pattern confirms that farmers respond to price variation in a way that is consistent with economic theory, and that demand for seed is indeed price-sensitive.

5.2 Three-stage pricing analysis

Having established that demand is price-sensitive, we now test whether the three pricing mechanisms—signaling, screening, and sunk cost—affect how farmers use the seed from the trial pack, whether they plan to use it in the future, and whether they adopt the promoted technology in subsequent seasons. We use regressions to estimate Equation 1 and isolate signaling, screening, and sunk cost effects of prices.

5.2.1 Trial pack use

We start by testing if, and how, farmers used the seed trial pack. Indeed, the primary outcome in studies that look at the relative importance of screening and sunk cost effects is whether the product that was subsidized or provided for free was used for its intended purpose and not wasted, diverted or sold. In the case of seed trial packs, however, use as intended goes beyond simply planting the seed. The purpose of a trial pack is to enable farmers to learn about a new variety; to observe its performance under local conditions and make an informed decision about whether to adopt it in subsequent seasons. Appropriate use therefore requires that farmers manage the trial in a way that facilitates such learning. This may also involve the use of complementary inputs and recommended agronomic practices, which help ensure that the trial provides an informative signal about the variety's performance.

Our first family of outcome variables—summarized in a Trial Seed Use Index following Anderson (2008)—is thus comprised of seven outcomes, all measured at midline, immediately after the season when farmers could try out the seed they were offered to buy. First, we simply check if the seed trial pack was used (that is, farmers reported they planted the seed on one for their fields). A second indicator of appropriate use is whether farmers kept the seed separate during planting. Planting the seed separately (rather than mixing it with local varieties) reflects a deliberate intention to evaluate the promoted variety on its own merits, which may influence adoption decisions in subsequent seasons. Similar to the case of planting, we also examine whether farmers kept the harvest from the seed trial pack separate. Doing so indicates that the variety's identity was maintained throughout the season, enabling an accurate assessment of its yield, market value, and other characteristics such as taste and appearance, and thus serves as an additional measure of appropriate use.

Another indicator of appropriate use concerns who actually planted the seed trial pack. Farmers who pay for the seed are likely to be more motivated and personally invested and may perceive the seed as higher quality, making them more inclined to plant it themselves rather than delegate the task to others. By contrast, those who receive it for free may feel less committed or confident in its value and thus be more willing to pass it on to a family member or hired laborer. Finally, we also measure whether farmers planted the seed correctly—using the recommended spacing and seed number per hole—as a direct indicator of careful engagement with the trial pack. In addition, we include two measures of complementary input use on the plot on which the trial pack was planted: whether the farmer applied fertilizer (organic or inorganic) and whether they used crop protection chemicals (pesticides, insecticides, or fungicides), as complementary inputs are important for the yield benefits of seed of improved varieties to materialize (Miehe et al., 2025), and their use reflects a deliberate investment in the trial.

Table 2 shows the results. Basic trial pack use is near universal in this sample: over 97 percent of farmers planted the seed, 94 percent kept the seed separate during planting, and almost 92 percent planted the seed themselves. This confirms that seed trial packs succeed at encouraging experimentation regardless of price.

Comparing treatment groups using the Trial Seed Use Index allows us to distinguish the three potential price mechanisms. We find evidence of screening: farmers with higher willingness to pay used the trial pack more appropriately, particularly by following recommended planting practices and applying fertilizer. We find no evidence of signaling effects. In contrast, we find a negative sunk cost effect: farmers who paid the full price (i.e., who did not receive the surprise discount) were less likely to use the trial pack in ways that facilitate

Table 2: Effects on Use of Trial Seed

	sample mean	signaling	screening	sunk cost	nobs
	Uganda - maize				
Trial Seed Use Index	0.076	-0.003	0.017**	-0.103**	669
	(0.334)	(0.011)	(0.006)	(0.037)	
Used trial pack as seed (1=yes)	0.972	-0.004	0.005	-0.014	749
	(0.165)	(0.004)	(0.003)	(0.014)	
Kept seed separate (1=yes)	0.94	-0.005	-0.002	-0.003	727
	(0.238)	(0.007)	(0.004)	(0.024)	
Kept harvest separate (1=yes)	0.713	0.007	0.004	-0.122**	683
	(0.452)	(0.016)	(0.009)	(0.047)	
Buyer used seed personally (1=yes)	0.918	-0.011	0.002	-0.044+	728
	(0.275)	(0.009)	(0.005)	(0.026)	
Correct planting (1=yes)	0.169	-0.011	0.016*	-0.035	714
	(0.375)	(0.011)	(0.006)	(0.036)	
Used fertilizer (1=yes)	0.546	0	0.021*	-0.078	728
	(0.498)	(0.014)	(0.01)	(0.055)	
Used pesticide (1=yes)	0.285	-0.009	0.01	-0.032	728
	(0.452)	(0.013)	(0.008)	(0.046)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

learning, primarily because they were less likely to keep the harvest separate for evaluation.

Appendix Table A1 presents an alternative specification in which the sunk cost effect is estimated using the transaction price—equal to the farmer's WTP price or zero if a full discount was received—instead of the binary indicator distinguishing payers from non-payers. The overall patterns are very similar to those reported in the main text, confirming the robustness of our findings: results show a significant screening effect on the Trial Seed Use Index, no significant signaling effects, and a negative sunk cost effect. Appendix Table A4 reports results after excluding right-censored observations, which yields the same pattern of findings.

5.2.2 Intentions

Beyond actual use of the seed trial pack, we also measured farmers' stated adoption intentions at midline (that is, before the next planting season in which actual adoption is assessed—see next section). These planning outcomes help bridge the gap between how farmers used the trial pack and what they subsequently did in the following season. We examine four outcomes: whether the farmer plans to use seed of an improved crop variety, such as a hybrid or OPV, in the next season; whether the farmer plans to use Bazooka specifically

Table 3: Effects on Intentions

	sample mean	signaling	screening	sunk cost	nobs
	Uganda - maize				
Adoption Intentions Index	0.282	-0.04^{+}	0.007	-0.036	473
	(0.54)	(0.021)	(0.012)	(0.051)	
Plans to use improved variety (1=yes)	0.627	0.009	0.025**	0.04	758
	(0.484)	(0.014)	(0.009)	(0.043)	
Plans to use Bazooka (1=yes)	0.581	0.007	0.023*	0.028	758
	(0.494)	(0.013)	(0.009)	(0.047)	
Planned area for improved variety (acres)	1.514	-0.066	0.019	0.149	473
	(1.421)	(0.046)	(0.023)	(0.156)	
Plans to buy seed (1=yes)	0.078	-0.011	0.005	-0.032	758
	(0.268)	(0.009)	(0.005)	(0.023)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

(the seed type used in the experiment); the planned area allocated to seed of improved crop varieties; and whether the farmer plans to purchase fresh seed from an agro-dealer. These measures are aggregated into a Adoption Intentions Index. Table 3 presents the results.

Screening has a positive effect on intentions: Farmers with higher willingness to pay for the seed trial pack are significantly more likely to plan to use seed of an improved maize variety in the next season and are also significantly more likely to plan to use the promoted Bazooka variety specifically. This pattern is consistent with the screening mechanism selecting farmers who are more motivated and better positioned to benefit from the technology. These farmers not only use the trial pack more carefully but also form stronger intentions to continue using improved varieties. We do not find evidence of sunk cost or signaling effects on intended use. Overall, these stated intentions mirror the screening patterns observed in use behavior, suggesting a behavioral pathway through which screening shapes learning, and intentions. Appendix Table A2 confirms the robustness of these results using the continuous transaction price specification; Appendix Table A5 shows similar results after discarding right-censored observations.

5.2.3 Adoption, input use, and production in subsequent season

Adoption of the promoted technology in the subsequent season is particularly relevant, as a primary aim of seed trial packs is to encourage experimentation that may lead to continued use. Observing behavior in the following season provides an early indication of whether farmers' trial experiences translate into repeated adoption.

The two adoption measures mirror the intention outcomes from the previous section but capture actual behavior on the randomly selected plot: whether seed of an improved variety was used (hybrid or OPV), and whether the promoted Bazooka variety was planted. We also examine whether farmers applied fertilizer or crop protection chemicals on the randomly selected plot, mirroring the complementary input measures from the trial pack season. Finally, we measure production and productivity on the randomly selected plot, both transformed using the inverse hyperbolic sine (IHS) to accommodate zero values while approximating a logarithmic transformation for larger values. As before, we aggregate outcomes into a Subsequent Season Index following Anderson (2008).

Results (in Table 4) reveal a nuanced pattern. Adoption measures show no significant treatment effects: neither willingness to pay (screening) nor the initial offer price (signaling) is associated with the likelihood of using improved or promoted seed in the following season, and complementary input use on the randomly selected plot does not differ across pricing arms. However, screening effects on production are significant and substantial: Farmers with higher willingness to pay achieved significantly higher production on the randomly selected plot in the subsequent season. Sunk cost effects on productivity are also significant but negative, mirroring the pattern observed during the trial season.

The combination of null adoption effects and significant production and productivity effects may be explained through seed recycling. Among farmers who grew the promoted Bazooka variety on the randomly selected plot in the subsequent season (69% of the sample), 83% used recycled seed from the harvested grain obtained from the seed trial pack, rather than freshly purchased certified seed (despite the fact that a hybrid variety such as Bazooka should not be reused). Recycling rates did not differ across treatment arms (84% among discounted versus 82% among full-price farmers, p=0.48), confirming that the production differences are not driven by differential adoption of the variety. Moreover, the screening effect on production remains significant when the sample is restricted to Bazooka recyclers (p=0.031), indicating that the effect operates within the group of farmers who recycle the seed trial pack variety.

The most likely channel through which pricing affects the quality of recycled seed is through first-season management practices. For the sunk cost channel specifically, farmers who paid for the trial pack saved significantly less grain as seed for the following season (p=0.039) and were less likely to keep the trial harvest separate from other varieties during the trial season (Table2), both of which would reduce the purity and quality of recycled seed. For the screening channel, higher willingness to pay selected farmers who managed the trial more carefully across multiple dimensions, including correct planting, complementary input use, and post-harvest separation. While we cannot isolate which specific management practice drives the subsequent production effect, the evidence is consistent with the view that pricing shapes first-season seed management in ways that affect the quality of recycled seed available for subsequent planting.

Appendix Table A3 shows that these patterns remain robust when using transaction prices instead of the binary payment indicator. The direction and significance of the coefficients are consistent across specifications, indicating that results are not driven by the binary distinction between paying and receiving

Table 4: Effects on Adoption, Input Use, and Production in Subsequent Season

	sample mean	signaling	screening	sunk cost	nobs
		Uganda - maize			
Subsequent Season Index	-0.004	0.001	0.009	-0.067	677
	(0.589)	(0.017)	(0.009)	(0.063)	
Used improved variety on random plot (1=yes)	0.218	0.012	0.006	0.029	703
	(0.413)	(0.012)	(0.008)	(0.043)	
Used promoted seed on random plot (1=yes)	0.113	0.003	0.005	0.014	703
	(0.317)	(0.009)	(0.006)	(0.031)	
Used fertilizer on random plot (1=yes)	0.511	-0.002	-0.002	-0.059	697
	(0.5)	(0.015)	(0.011)	(0.053)	
Used chemicals on random plot (1=yes)	0.401	0.008	-0.005	-0.032	698
	(0.491)	(0.013)	(0.009)	(0.049)	
Production (IHS)	6.139	-0.031	0.06**	-0.147	686
	(1.113)	(0.033)	(0.02)	(0.111)	
Productivity (IHS)	6.487	-0.035	0.029	-0.223*	678
	(1.068)	(0.033)	(0.018)	(0.095)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

the seed for free. Furthermore, Appendix Table A6 repeats the analysis for non-censored observations, generating similar findings.

6 Conclusion

In this paper, we examined how pricing influences the introduction of a new technology, focusing on an improved maize variety distributed to smallholder farmers in Uganda. Temporary discounts are often justified as a way to encourage trial of an unfamiliar product, but critics warn that free provision can lead to waste or misuse. We tested whether charging a positive price affects use and adoption through three distinct mechanisms. First, low or zero prices may signal that the product is ineffective, of poor quality, or of limited usefulness, thereby reducing the effort invested in its use. Second, prices can act as a screening device, allocating products to those who value them most. Third, paying for a product may strengthen commitment to use, consistent with the sunk cost rationale in behavioral economics.

We identify these mechanisms through a field experiment using a three-stage pricing design that builds on and extends the typical two-stage designs used to study screening and sunk cost effects. In the first stage, a seed trial pack was offered at an initial (randomized) price that could serve as a signal for unobservable attributes of the trial pack. In the second stage, farmers bargained over the price they would pay for the trial pack through a structured bargaining process that allowed the negotiated price to converge toward a proxy for the farmer's willingness to pay. In the third stage, a randomly selected subset of farmers received a surprise discount on the price negotiated in the previous stage, lowering the amount they ultimately paid while leaving the negotiated price unchanged. This structure generated three distinct price measures that allowed us to separately estimate signaling, screening, and sunk cost effects.

We tested whether these prices influenced trial pack use, adoption intentions, subsequent adoption of the promoted technology, and production. Overall, willingness to pay was associated with more appropriate use of the trial seed and greater investment in complementary inputs during the trial season, consistent with a screening effect. Willingness to pay also predicted stronger intentions to adopt improved varieties in the following season, including the promoted Bazooka variety. Initial high offer prices did not serve as a signal for attributes such as effectiveness or quality. Sunk cost effects were negative, suggesting that paying for the seed reduced rather than reinforced appropriate use. While adoption rates and input use in the subsequent season did not differ across treatment arms, screening effects on production persisted; since most farmers recycled seed from their trial harvest, pricing-induced differences in first-season management appear to have affected the quality of recycled seed available for subsequent planting.

The results point to several mechanisms that may explain how prices influenced farmers' behavior in our setting. First, signaling effects were weak. Because the seeds were distributed by enumerators affiliated with a research organization, farmers may have assumed the technology was appropriate for them and trusted the quality regardless of the stated price. In addition, trial packs were low-risk, and the price variation may have been too small or not credible enough to meaningfully signal quality. In markets where sellers have established reputations or products carry certification, or where farmers hold strong prior beliefs about what works, prices may carry little additional informational content about less visible product attributes.

Screening effects were positive and significant, suggesting that farmers' willingness to pay reflects underlying differences in capacity or motivation. Farmers who were willing to pay more for the trial seed tended to invest more effort in proper planting, management, and complementary inputs, indicating that price can serve as a signal of readiness to adopt new practices. These behavioral differences likely had downstream consequences: because most farmers recycled seed from the trial harvest rather than purchasing new certified seed, better-managed trial plots generated higher-quality recycled seed for the following season, contributing to higher production among higher-paying farmers. In other words, willingness to pay appears to screen for farmers who are both willing and able to use the technology effectively, amplifying use and productivity through both immediate management and longer-term effects on seed quality.

Sunk cost effects were negative. Also here, several mechanisms could explain why paying reduced appropriate use in our setting. Free provision may have evoked a gift-exchange framing, encouraging experimentation, while payment framed the seed as a routine market transaction, reducing motivation to observe and learn. Relatedly, farmers who received the seed for free may have treated it as a windfall, making them more willing to experiment and invest effort because the downside of failure felt costless (a "house money" effect, as in Thaler and Johnson 1990). Payment may also have shifted behavior from experimentation toward risk avoidance, deterring farmers from keeping the harvest separate and evaluating the variety carefully. Finally, while the strongest sunk cost effects appear to operate through reduced experimentation and careful observation, the fact that complementary input use is also directionally negative suggests that liquidity or credit constraints may have played a role as well, which is in line with suggestive evidence in Mahmoud (2024).

The findings provide several insights for the design of seed trial pack programs aimed at promoting improved agricultural technologies. In the study context, higher prices did not convey information about less observable product attributes, such as effectiveness or quality. Farmers appeared to trust the seeds provided by research-affiliated enumerators regardless of price, suggesting that confidence-building efforts may be more effective through alternative channels, including certification, branding, demonstrations, and trusted agro-dealer networks rather than through pricing alone (see also Bulte et al. 2014; BenYishay and Mobarak 2019).

Regarding screening effects, offering trial seeds at a modest price can help identify farmers with the capacity and motivation to adopt new technologies effectively, ensuring that program resources are concentrated on those most likely to manage the trial well (Ashraf et al. 2010; Berry et al. 2020). Higher-paying farmers, who invest more effort and generate better-quality recycled seed, can also serve as local demonstration or "lead farmer" models, turning their superior management into a community learning resource (Kondylis et al., 2017; Emerick and Dar, 2021). Follow-on incentives, such as access to inputs, technical support, or recognition, can further encourage careful trial management and sustained adoption, ensuring that the screening function of price translates into lasting productivity and learning benefits (Buehren et al. 2025).

At the same time, sunk cost effects highlight that payment can reduce appropriate use of the trial seed. Distributing small trial packs at low or no cost can encourage experimentation by reducing perceived risk and leveraging "house money" and gift-exchange effects (Cohen and Dupas 2010; Omotilewa et al. 2019). Pairing trials with complementary input support or short-term credit can mitigate liquidity constraints and ensure proper implementation, while structured follow-up activities, such as field demonstrations, peer learning sessions, or simple observation guides, can help farmers capture the full informational value of the trial (Hanna et al. 2014).

Taken together, these findings underscore that prices play multiple, sometimes opposing, roles in shaping technology adoption, and that their effects are highly context-dependent. Program designers should consider whether the goal is to screen for motivated adopters or to maximize experimentation and learning, and tailor pricing, support, and follow-up strategies accordingly. As such, prices are not merely a tool for cost recovery; they can be leveraged deliberately to accelerate learning, adoption, and productivity.

Ethical clearance

This research received clearance from Makerere's School of Social Sciences Research Ethics Committee (MAKSSREC 01.23.627/PR1) as well as from IFPRI IRB (DSGD-23-0108). The research was also registered at the Ugandan National Commission for Science and Technology (SS1657ES).

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data and code used in this study are available at the project GitHub repository.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT in order to refine phrasing, improve clarity and structure, and prepare materials for publication and dissemination. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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A Appendix: Estimates Based on the Continuous Transaction Price

Table A1: Effects on Use of Trial Seed

	sample mean	signaling	screening	sunk cost	nobs
		Uganda - maize			
Trial Seed Use Index	0.076	0.001	0.025**	-0.014*	669
	(0.334)	(0.011)	(0.006)	(0.006)	
Used trial pack as seed (1=yes)	0.972	-0.003	0.006+	-0.002	749
	(0.165)	(0.004)	(0.003)	(0.002)	
Kept seed separate (1=yes)	0.94	-0.005	-0.002	0.001	727
	(0.238)	(0.007)	(0.005)	(0.004)	
Kept harvest separate (1=yes)	0.713	0.007	0.013	-0.018*	683
	(0.452)	(0.016)	(0.009)	(0.007)	
Buyer used seed personally (1=yes)	0.918	-0.01	0.005	-0.006	728
	(0.275)	(0.01)	(0.005)	(0.004)	
Correct planting (1=yes)	0.169	-0.013	0.019**	-0.007	714
	(0.375)	(0.012)	(0.006)	(0.006)	
Used fertilizer (1=yes)	0.546	0.01	0.027**	-0.007	728
	(0.498)	(0.014)	(0.01)	(0.009)	
Used pesticide (1=yes)	0.285	-0.005	0.013	-0.002	728
	(0.452)	(0.013)	(0.009)	(0.007)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

Table A2: Effects on Intentions

	sample mean	signaling	screening	sunk cost	nobs
		Uganda - maize			
Adoption Intentions Index	0.282	-0.042*	0.01	-0.005	473
	(0.54)	(0.02)	(0.012)	(0.009)	
Plans to use improved variety (1=yes)	0.627	0.008	0.022*	0.007	758
	(0.484)	(0.014)	(0.01)	(0.007)	
Plans to use Bazooka (1=yes)	0.581	0.008	0.02*	0.005	758
	(0.494)	(0.014)	(0.01)	(0.007)	
Planned area for improved variety (acres)	1.514	-0.076	0.006	0.031	473
	(1.421)	(0.052)	(0.025)	(0.026)	
Plans to buy seed (1=yes)	0.078	-0.012	0.007	-0.006	758
	(0.268)	(0.009)	(0.005)	(0.004)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

Table A3: Effects on Adoption, Input Use, and Production in Subsequent Season

	sample mean	signaling	screening	sunk cost	nobs
			Uganda - maize		
Subsequent Season Index	-0.004	0.012	0.015	-0.008	677
	(0.589)	(0.019)	(0.01)	(0.01)	
Used improved variety on random plot (1=yes)	0.218	0.011	0.004	0.005	703
	(0.413)	(0.013)	(0.008)	(0.007)	
Used promoted seed on random plot (1=yes)	0.113	0.007	0.004	0.002	703
	(0.317)	(0.009)	(0.006)	(0.005)	
Used fertilizer on random plot (1=yes)	0.511	0.003	0.002	-0.006	697
	(0.5)	(0.017)	(0.012)	(0.008)	
Used chemicals on random plot (1=yes)	0.401	0.015	-0.002	-0.003	698
	(0.491)	(0.014)	(0.01)	(0.008)	
Production (IHS)	6.139	-0.017	0.074**	-0.022	686
	(1.113)	(0.036)	(0.023)	(0.019)	
Productivity (IHS)	6.487	-0.024	0.048*	-0.036*	678
	(1.068)	(0.035)	(0.02)	(0.016)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels.

B Appendix: Excluding Right-Censored Observations

As a robustness check, we re-estimate all main specifications after excluding farmers whose willingness to pay was right-censored---that is, those who accepted the initial offer price without negotiation. For these 111 farmers, the observed transaction price represents a lower bound on their true willingness to pay, potentially attenuating the screening estimates. Tables A4--A6 report the results. The main findings are qualitatively unchanged. Screening effects on trial seed use remain significant. The pattern of null adoption and input use results combined with significant production effects in the subsequent season is confirmed.

Table A4: Effects on Use of Trial Seed (Excluding Right-Censored)

	sample mean	signaling	screening	sunk cost	nobs
	Uganda - maize				
Trial Seed Use Index	0.076	-0.003	0.017**	-0.103**	669
	(0.334)	(0.011)	(0.006)	(0.037)	
Used trial pack as seed (1=yes)	0.972	-0.004	0.005	-0.014	749
	(0.165)	(0.004)	(0.003)	(0.014)	
Kept seed separate (1=yes)	0.94	-0.005	-0.002	-0.003	727
	(0.238)	(0.007)	(0.004)	(0.024)	
Kept harvest separate (1=yes)	0.713	0.007	0.004	-0.122**	683
	(0.452)	(0.016)	(0.009)	(0.047)	
Buyer used seed personally (1=yes)	0.918	-0.011	0.002	-0.044+	728
	(0.275)	(0.009)	(0.005)	(0.026)	
Correct planting (1=yes)	0.169	-0.011	0.016*	-0.035	714
	(0.375)	(0.011)	(0.006)	(0.036)	
Used fertilizer (1=yes)	0.546	0	0.021*	-0.078	728
	(0.498)	(0.014)	(0.01)	(0.055)	
Used pesticide (1=yes)	0.285	-0.009	0.01	-0.032	728
	(0.452)	(0.013)	(0.008)	(0.046)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels. Sample excludes 118 farmers whose willingness to pay was right-censored (accepted initial offer without negotiation).

Table A5: Effects on Intentions (Excluding Right-Censored)

	sample mean	signaling	screening	sunk cost	nobs
	Uganda - maize				
Adoption Intentions Index	0.282	-0.04+	0.007	-0.036	473
	(0.54)	(0.021)	(0.012)	(0.051)	
Plans to use improved variety (1=yes)	0.627	0.009	0.025**	0.04	758
	(0.484)	(0.014)	(0.009)	(0.043)	
Plans to use Bazooka (1=yes)	0.581	0.007	0.023*	0.028	758
	(0.494)	(0.013)	(0.009)	(0.047)	
Planned area for improved variety (acres)	1.514	-0.066	0.019	0.149	473
	(1.421)	(0.046)	(0.023)	(0.156)	
Plans to buy seed (1=yes)	0.078	-0.011	0.005	-0.032	758
	(0.268)	(0.009)	(0.005)	(0.023)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels. Sample excludes 118 farmers whose willingness to pay was right-censored (accepted initial offer without negotiation).

Table A6: Effects on Adoption, Input Use, and Production in Subsequent Season (Excluding Right-Censored)

	sample mean	signaling	screening	sunk cost	nobs
	Uganda - maize				
Subsequent Season Index	-0.004	0.001	0.009	-0.067	677
	(0.589)	(0.017)	(0.009)	(0.063)	
Used improved variety on random plot (1=yes)	0.218	0.012	0.006	0.029	703
	(0.413)	(0.012)	(0.008)	(0.043)	
Used promoted seed on random plot (1=yes)	0.113	0.003	0.005	0.014	703
	(0.317)	(0.009)	(0.006)	(0.031)	
Used fertilizer on random plot (1=yes)	0.511	-0.002	-0.002	-0.059	697
	(0.5)	(0.015)	(0.011)	(0.053)	
Used chemicals on random plot (1=yes)	0.401	0.008	-0.005	-0.032	698
	(0.491)	(0.013)	(0.009)	(0.049)	
Production (IHS)	6.139	-0.031	0.06**	-0.147	686
	(1.113)	(0.033)	(0.02)	(0.111)	
Productivity (IHS)	6.487	-0.035	0.029	-0.223*	678
	(1.068)	(0.033)	(0.018)	(0.095)	

Note: Village-clustered standard errors in parentheses. **, *, and + denote significance at the 1, 5, and 10% levels. Sample excludes 118 farmers whose willingness to pay was right-censored (accepted initial offer without negotiation).